

Best -Known Semiconductor Properties

	Si	Ge	C	AlN	GaN		InN	InP	GaAs	InAs	GaSb	InSb
					Wurtzite	Zinc Blende						
Bandgap (eV)	1.1	0.67	5.47	6.2	3.39	3.28	1.9 – 0.7 (Unknown)	1.35	1.424	0.354	0.726	0.17
Lattice Constant (nm)	0.54	0.57	0.36	0.32	0.52	0.32	0.36	0.586	0.565	0.606	0.61	0.65
Electron mobility (cm ² /Vs)	1350	3900	2800	300	=<1000	=<1000	3200	5400	8500	40000	3000	77000
Hole mobility (cm ² /Vs)	480	1900	2010	14	=<200	=<350	UNKNOWN	200	200	500	1000	850
Electron Saturation Velocity (x10 ⁷ cm/s)	1	0.6	?	--	2.8	3	4	4	0.7	6	4	5
Thermal conductivity (W/cm-K)	1.3	0.58	25	3.5	1.3	1.3	0.45	0.68	0.55	0.27	0.32	0.18
Relative dielectric	11.7	16.2	5.7	8.5	8.9	9.7	15.3	12.5	12.9	15.15	15.7	16.8
Electron Effective Mass	0.26	0.22	?	0.4	0.2	0.13	0.11	0.08	0.063	0.41		0.43
Conduction Band Density (x10 ¹⁹)cm ⁻³	3.2	1	?	0.63	0.23	0.12	0.09	0.057	0.047	0.0087	0.021	0.0042
Valley Energy Gap (eV)	0.9	0.65	?	NA	1.4	1.4	3.9	0.6	0.48	0.354	0.05	0.5
Hole Effective Mass	0.49	0.33	?	7.26	0.19	0.33	1.63	0.6	0.51	0.026	0.4	0.43
Valence Band Density (x10 ¹⁹)cm ⁻³	1.8	0.5	?	48	4.1	4.6	5.3	1.1	0.09	0.66	1.8	0.73

<http://www.ioffe.ru/SVA/NSM/Semicond/InAs/basic.html>